

```
from google.colab import drive
drive.mount('/content/drive/', force_remount = True)
```

Mounted at /content/drive/

```
#Packages installation
```

```
!pip install SentencePiece
```

```
!pip install transformers
```

```
!pip install googletrans==3.1.0a0
```

```
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.10/dist-packages (from huggingface)
Collecting huggingface-hub<1.0,>=0.16.4 (from transformers)
  Downloading huggingface_hub-0.17.3-py3-none-any.whl (295 kB)
    295.0/295.0 kB 26.0 MB/s eta 0:00:00
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests->tra
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests->transformers) (
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests->transform
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests->transform
Installing collected packages: safetensors, huggingface-hub, tokenizers, transformers
Successfully installed huggingface-hub-0.17.3 safetensors-0.4.0 tokenizers-0.14.1 transformers-4.35.0
Collecting googletrans==3.1.0a0
  Downloading googletrans-3.1.0a0.tar.gz (19 kB)
  Preparing metadata (setup.py) ... done
Collecting httpx==0.13.3 (from googletrans==3.1.0a0)
  Downloading httpx-0.13.3-py3-none-any.whl (55 kB)
    55.1/55.1 kB 1.7 MB/s eta 0:00:00
Requirement already satisfied: certifi in /usr/local/lib/python3.10/dist-packages (from httpx==0.13.3->googletrans==3.
Collecting hstspreload (from httpx==0.13.3->googletrans==3.1.0a0)
  Downloading hstspreload-2023.1.1-py3-none-any.whl (1.5 MB)
    1.5/1.5 MB 19.7 MB/s eta 0:00:00
Requirement already satisfied: sniffio in /usr/local/lib/python3.10/dist-packages (from httpx==0.13.3->googletrans==3.
Collecting chardet==3.* (from httpx==0.13.3->googletrans==3.1.0a0)
  Downloading chardet-3.0.4-py2.py3-none-any.whl (133 kB)
    133.4/133.4 kB 17.3 MB/s eta 0:00:00
Collecting idna==2.* (from httpx==0.13.3->googletrans==3.1.0a0)
  Downloading idna-2.10-py2.py3-none-any.whl (58 kB)
    58.8/58.8 kB 7.0 MB/s eta 0:00:00
Collecting rfc3986<2,>=1.3 (from httpx==0.13.3->googletrans==3.1.0a0)
  Downloading rfc3986-1.5.0-py2.py3-none-any.whl (31 kB)
Collecting httpcore==0.9.* (from httpx==0.13.3->googletrans==3.1.0a0)
  Downloading httpcore-0.9.1-py3-none-any.whl (42 kB)
    42.6/42.6 kB 4.6 MB/s eta 0:00:00
Collecting h11<0.10,>=0.8 (from httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
  Downloading h11-0.9.0-py2.py3-none-any.whl (53 kB)
    53.6/53.6 kB 6.8 MB/s eta 0:00:00
Collecting h2==3.* (from httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
  Downloading h2-3.2.0-py2.py3-none-any.whl (65 kB)
    65.0/65.0 kB 8.1 MB/s eta 0:00:00
Collecting hyperframe<6,>=5.2.0 (from h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
  Downloading hyperframe-5.2.0-py2.py3-none-any.whl (12 kB)
Collecting hpack<4,>=3.0 (from h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==3.1.0a0)
  Downloading hpack-3.0.0-py2.py3-none-any.whl (38 kB)
Building wheels for collected packages: googletrans
  Building wheel for googletrans (setup.py) ... done
  Created wheel for googletrans: filename=googletrans-3.1.0a0-py3-none-any.whl size=16352 sha256=76154168d9ec7c216f32f
  Stored in directory: /root/.cache/pip/wheels/50/5d/3c/8477d0af4ca2b8b1308812c09f1930863caeabc762fe265a95
Successfully built googletrans
Installing collected packages: rfc3986, hyperframe, hpack, h11, chardet, idna, hstspreload, h2, httpcore, httpx, googl
  Attempting uninstall: chardet
    Found existing installation: chardet 5.2.0
    Uninstalling chardet-5.2.0:
      Successfully uninstalled chardet-5.2.0
  Attempting uninstall: idna
    Found existing installation: idna 3.4
    Uninstalling idna-3.4:
      Successfully uninstalled idna-3.4
Successfully installed chardet-3.0.4 googletrans-3.1.0a0 h11-0.9.0 h2-3.2.0 hpack-3.0.0 hstspreload-2023.1.1 httpcore-
```

```
# import libraries
```

```
import os
```

```
import numpy as np
```

```
import pandas as pd
```

```
import googletrans
```

```
from googletrans import Translator
```

```
import torch
```

```
from transformers import DistilBertTokenizer, DistilBertForSequenceClassification, pipeline
```

```
#import excel file from google drive
```

```
df = pd.read_excel('/content/drive/MyDrive/Freelancing/NLP/From GL/sentiment_analysis_EXTERNS.2023-10-27.xlsx')
```

```
df
```



|     | id           | text_answer                                         | text_length |
|-----|--------------|-----------------------------------------------------|-------------|
| 0   | ext_f3b0f7ee | Auf mein Spezialgebiet Knie wurde nur oberfläch...  | 63          |
| 1   | ext_e4740034 | Leider waren alle eingearbeiteten Empfehlungen...   | 539         |
| 2   | ext_12ed912c | Hervorragendes Format! Ich danke Ihnen sehr he...   | 156         |
| 3   | ext_b74c9803 | Mein tatsächlicher Tätigkeitsschwerpunkt wird du... | 167         |
| 4   | ext_c7a20686 | Ich schätze Ihre Arbeit sehr und bin frei von l...  | 83          |
| ... | ...          | ...                                                 | ...         |
| 769 | ext_2bac289a | keine anmerkungen                                   | 17          |
| 770 | ext_2cf3ec97 | Das innovativ strukturierte Aufgabenspektrum m...   | 173         |
| 771 | ext_39c815d5 | Schlau wäre es, wenn Sie jeden Arzt bitten, jew...  | 190         |
| 772 | ext_882ea547 | Unterpunkt Kinderwirbelsäulenchirurgie wäre noch... | 150         |
| 773 | ext_b03d6a8c | Als medizinische Vertiefungsthemen wurden folg...   | 156         |

774 rows x 3 columns

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

▼ Translating Surveys From German language into English

```
# Define function
def translate(text):
    translator = Translator()
    return translator.translate(text,src='de', dest='en').text

df['English Translated Text'] = df['text_answer'].apply(translate)
df
```



|     | id           | text_answer                                         | text_length | English Translated Text                           |
|-----|--------------|-----------------------------------------------------|-------------|---------------------------------------------------|
| 0   | ext_f3b0f7ee | Auf mein Spezialgebiet Knie wurde nur oberfläch...  | 63          | My specialty, the knee, was only discussed sup... |
| 1   | ext_e4740034 | Leider waren alle eingearbeiteten Empfehlungen...   | 539         | Unfortunately, when the survey was called up a... |
| 2   | ext_12ed912c | Hervorragendes Format! Ich danke Ihnen sehr he...   | 156         | Excellent format! I would like to thank you ve... |
| 3   | ext_b74c9803 | Mein tatsächlicher Tätigkeitsschwerpunkt wird du... | 167         | My actual focus of activity is not captured by... |
| 4   | ext_c7a20686 | Ich schätze Ihre Arbeit sehr und bin frei von l...  | 83          | I value your work very much and am free from c... |
| ... | ...          | ...                                                 | ...         | ...                                               |
| 769 | ext_2bac289a | keine anmerkungen                                   | 17          | no comments                                       |

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

▼ Assigning Sentiments to the texts using Distilbert Transformers

Huggingface model: [Click Here](#)

Zum Bearbeiten doppelklicken (oder Eingabe)

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
tokenizer = DistilBertTokenizer.from_pretrained("distilbert-base-uncased") # Target variable has been assigned by fine tuned
model = DistilBertForSequenceClassification.from_pretrained("distilbert-base-uncased-finetuned-sst-2-english")
```

```

Downloading (...)okenizer_config.json: 28.0/28.0 [00:00<00:00,
100% 879B/s]

Downloading (...)solve/main/vocab.txt: 100% 593kB/s]

Downloading (...)main/tokenizer.json: 100% 794kB/s]

Downloading (...)lve/main/config.json: 483/483 [00:00<00:00,

```

```
sentiments = []
```

```
for i in df["English Translated Text"]:
```

```

try:
    inputs = tokenizer(i, return_tensors="pt")

    with torch.no_grad():
        logits = model(**inputs).logits

    predicted_class_id = logits.argmax().item()
    sentiments.append(model.config.id2label[predicted_class_id])

```

```

except:
    sentiments.append("Null")

```

Token indices sequence length is longer than the specified maximum sequence length for this model (1171 > 512). Running

```
len(sentiments)
```

```
774
```

```

# Adding column sentiments
df['Text Sentiments'] = sentiments

```

```
df
```

|   | Unnamed: 0 | id           | text_answer                                        | text_length | English Translated Text                           | Senti |
|---|------------|--------------|----------------------------------------------------|-------------|---------------------------------------------------|-------|
| 0 | 0          | ext_f3b0f7ee | Auf mein Spezialgebiet Knie wurde nur oberfläch... | 63          | My specialty, the knee, was only discussed sup... | NEC   |
| 1 | 1          | ext_e4740034 | Leider waren alle eingearbeiteten Empfehlungen...  | 539         | Unfortunately, when the survey was called up a... | NEC   |
| 2 | 2          | ext_12ed912c | Hervorragendes Format! Ich danke Ihnen sehr he...  | 156         | Excellent format! Thank you very much for the ... | PC    |
| 3 | 3          | ext_b74c9803 | Mein tatsächlicher Tätigkeitsschwerpunkt wird      | 167         | My actual focus of activity is not                | NEC   |

```

#Saving the data with sentiments to an excel file
df.to_excel("english_translated_texts_with_sentiments.xlsx")

```

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

## ✓ Finding the answers using Roberta Transformers

# how you put only 'here' and it directs you directly to the link

Huggingface model [here](#)

```
# any work or any file showing that you cleaned some data manually
```

**Note: I manually cleaned the data a bit like "..." or "---" or ":" etc**

```
context = "" #where does it come from
```

```
for i in df['English Translated Text']:
    context = f""""{context}
    {i}""""
```

```
context
```

```
↳ '\n My specialty, the knee, was only discussed superficially.\n Unfortunately, when the survey was called up again for further processing, all of the recommendations that had been incorporated into the recommendation list were deleted again, meaning that all the information had to be entered again. Furthermore, it would also be helpful and more time-effective if the information in a field, e.g. B. "Further information on this medical in-depth topic" was entered from the previous year's survey, but would still be available for a simple revision if it were currently being edited. Everything had to be written down/re-entered here and it took time.\n Excellent format! Thank you very much
```

```
# print(context) # it will show the text line by line
```

```
# Perform question-answering for each question
```

```
questions = [
    "Why are people unhappy?",
    "What to do to improve our survey?",
    "What is good in the survey?",
    "Where are the interesting feedback and not just a comment like 'Cool'"
]
```

```
answer = pipeline("question-answering", model="consciousAI/question-answering-roberta-base-s-v2")
```

```
#Google tapas, roberta base-s-v2, tried different versions of roberta
```

```
#Please suggest if you have any preference
```

```
↳ Downloading (...)lve/main/config.json: 782/782 [00:00<00:00,
100% 19.4kB/s]
Downloading model.safetensors: 496M/496M [00:04<00:00,
100% 177MB/s]
Downloading (...)okenizer_config.json: 100% 1.58k/1.58k [00:00<00:00,
31.2kB/s]
Downloading (...)olve/main/vocab.json: 100% 798k/798k [00:00<00:00,
4.59MB/s]
```

```
# Summarization model
```

```
a = []
for question in questions:
    print(f"Question: {question}")
    print()
    print(f"Answer: {answer(question=question, context=context)}")
    print()
    a.append(answer(question=question, context=context))
    print("=====")
    print()
```

```
↳ Question: Why are people unhappy?
```

```
Answer: {'score': 0.8945690393447876, 'start': 30968, 'end': 31014, 'answer': 'Many doctors today work at different loca
=====
```

```
Question: What to do to improve our survey?
```

```
Answer: {'score': 0.6555940508842468, 'start': 107945, 'end': 107994, 'answer': 'The questionnaire has become much more
=====
```

```
Question: What is good in the survey?
```

```
Answer: {'score': 0.874576210975647, 'start': 78098, 'end': 78126, 'answer': 'Good and clear questionnaire'}
=====
```

Question: Where are the interesting feedback and not just a comment like 'Cool'

Answer: {'score': 0.651715874671936, 'start': 133393, 'end': 133448, 'answer': 'Feedback to us is provided by our special

=====

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
a = []
for question in questions:
    print(f"Question: {question}")
    print()
    ans = answer(question=question, context=context)
    print(f"Answer: {ans}")
    print()
    a.append(ans['answer'])
    print("=====")
    print()
```

➦ Question: Why are people unhappy?

Answer: {'score': 0.8945690393447876, 'start': 30968, 'end': 31014, 'answer': 'Many doctors today work at different loca

=====

Question: What to do to improve our survey?

Answer: {'score': 0.6555940508842468, 'start': 107945, 'end': 107994, 'answer': 'The questionnaire has become much more

=====

Question: What is good in the survey?

Answer: {'score': 0.874576210975647, 'start': 78098, 'end': 78126, 'answer': 'Good and clear questionnaire'}

=====

Question: Where are the interesting feedback and not just a comment like 'Cool'

Answer: {'score': 0.651715874671936, 'start': 133393, 'end': 133448, 'answer': 'Feedback to us is provided by our special

=====

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

a

➦ ['Many doctors today work at different locations',  
'The questionnaire has become much more practical.',  
'Good and clear questionnaire',  
'Feedback to us is provided by our specialist colleagues']

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
data = {'Questions': questions, 'Answers': a}
df = pd.DataFrame.from_dict(data)
df
```

➦

|   | Questions                                  | Answers                                                 |
|---|--------------------------------------------|---------------------------------------------------------|
| 0 | Why are people unhappy?                    | Many doctors today work at different locations          |
| 1 | What to do to improve our survey?          | The questionnaire has become much more practical.       |
| 2 | What is good in the survey?                | Good and clear questionnaire                            |
| 3 | Where are the interesting feedback and not | Feedback to us is provided by our specialist colleagues |

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
def translate_eng_2_ger(text):
    translator = Translator()
    return translator.translate(text, src='en', dest='de').text

df['German Translated Text'] = df['Answers'].apply(translate_eng_2_ger)
df
```



|   | Questions                         | Answers                                           | German Translated Text                            |
|---|-----------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0 | Why are people unhappy?           | Many doctors today work at different locations    | Viele Ärzte arbeiten heute an verschiedenen St... |
| 1 | What to do to improve our survey? | The questionnaire has become much more practical. | Der Fragebogen ist viel praktischer geworden.     |
| 2 | What is good in the survey?       | Good and clear questionnaire                      | Guter und klarer Fragebogen                       |

```
df.to_excel("Survey_QA.xlsx")
```

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

## ✓ Prompting in Loop

type **"DONE"** as input to break the loop

```
from transformers import pipeline #why done is used
```

```
context = ""
```

```
for i in df['English Translated Text']:
    context = f""""{context}
    {i}""""
```

```
answer = pipeline("question-answering", model="consciousAI/question-answering-roberta-base-s-v2")
```

```
while True:
```

```
    question = input("What is your question?: ")
```

```
    if question == "DONE":
        break
```

```
    else:
        print()
        print(f"Question: {question}")
        print()
        ans = answer(question=question, context=context)
        print(f"Answer: {ans}")
        print()
        print("=====")
        print()
```



```
What is your question?: Was this survey helpful?
```

```
Question: Was this survey helpful?
```

```
Answer: {'score': 0.7819448709487915, 'start': 75318, 'end': 75348, 'answer': 'The questionnaire is very good'}
```

```
=====
```

```
What is your question?: DONE
```

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.